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LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error".

```
5 void error(String msg);
6 }
7 public class MyLogger implements Logger {
8     public void debug(String msg) {
9         System.out.println("DEBUG: " + msg);
10    }
11    public void info(String msg) {
12        System.out.println("INFO: " + msg);
13    }
14    public void warning(String msg) {
15        System.out.println("WARNING: " + msg);
16    }
17    public void error(String msg) {
18        System.out.println("ERROR: " + msg);
19    }
20    public static void main(String[] args) {
21        MyLogger log = new MyLogger();
22        log.debug("Program started");
23        log.info("Data loaded");
24    }
25 }
```

OUTPUT

DEBUG: Program started
INFO: Data loaded
WARNING: Low memory
ERROR: File not found

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CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement

```
3 void attack();
4 void takeDamage();
5 }
6 public class CharacterDemo implements Character {
7     public void move() {
8         System.out.println("Character is moving");
9     }
10    public void attack() {
11        System.out.println("Character is attacking");
12    }
13    public void takeDamage() {
14        System.out.println("Character takes damage");
15    }
16    public static void main(String[] args) {
17        CharacterDemo c = new CharacterDemo();
18        c.move();
19        c.attack();
20        c.takeDamage();
21    }
22 }
```

OUTPUT

Character is moving
Character is attacking
Character takes damage

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⌵

Sort

Sort

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

```
1 import java.util.*;
2 public class Sort {
3     public static void main(String[] args) {
4         ArrayList<Integer> list = new ArrayList<>();
5         list.add(50);
6         list.add(20);
7         list.add(40);
8         list.add(10);
9         list.add(30);
10        Collections.sort(list);
11        System.out.println(list);
12    }
13 }
```

OUTPUT

[10, 20, 30, 40, 50]

Your INPUT goes here!

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⌵

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

```
1 import java.util.*;
2 public class HashSet {
3     public static void main(String[] args) {
4         java.util.HashSet<String> set = new java.util.HashSet<>();
5         set.add("Apple");
6         set.add("Banana");
7         set.add("Orange");
8         System.out.println(set);
9     }
10 }
```

OUTPUT

[Apple, Orange, Banana]

Your INPUT goes here!

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TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

```
2 public class TreeMap {
3     public static void main(String[] args) {
4         java.util.TreeMap<String, Integer> map = new java.util.TreeMap<>();
5         map.put("Apple", 10);
6         map.put("Banana", 20);
7         map.put("Orange", 30);
8         for (Map.Entry<String, Integer> entry : map.entrySet()) {
9             System.out.println(entry.getKey() + " = " + entry.getValue());
10        }
11    }
12 }
```

OUTPUT

```
Apple = 10
Banana = 20
Orange = 30
```

here!

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PriorityQueue ▾

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

```
1 import java.util.*;
2 public class PriorityQueue {
3     public static void main(String[] args) {
4         java.util.PriorityQueue<Integer> pq = new java.util.PriorityQueue<>();
5         pq.add(30);
6         pq.add(10);
7         pq.add(20);
8         pq.add(40);
9         System.out.println(pq);
10    }
11 }
```

OUTPUT

```
[10, 30, 20, 40]
```

Your INPUT goes here!

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⌵

LINKEDHASHMAP
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Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

```
2 public class LinkedHashMap {
3     public static void main(String[] args) {
4         java.util.LinkedHashMap<String, Integer> map = new java.util.LinkedHashMap<>();
5         map.put("Apple", 10);
6         map.put("Banana", 20);
7         map.put("Orange", 30);
8         for (String key : map.keySet()) {
9             System.out.println(key);
10        }
11    }
12 }
```

OUTPUT

Apple
Banana
Orange

here!

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⌵

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

```
1 import java.util.*;
2 public class MaxMin {
3     public static void main(String[] args) {
4         Set<Integer> set = new HashSet<>();
5         set.add(40);
6         set.add(10);
7         set.add(30);
8         set.add(20);
9         System.out.println("Maximum = " + Collections.max(set));
10        System.out.println("Minimum = " + Collections.min(set));
11    }
12 }
```

OUTPUT

Maximum = 40
Minimum = 10

Your INPUT goes here!